Chapter 3

General results of representation theory

3.1. Subrepresentations in semisimple representations

Let A be an algebra over an algebraically closed field k.

Definition 3.1.1. A **semisimple** (or **completely reducible**) representation of A is a direct sum of irreducible representations.

Example 3.1.2. Let V be an irreducible representation of A of dimension n. Then $Y = \mathrm{End}(V)$, with action of A by left multiplication, is a semisimple representation of A, isomorphic to nV (the direct sum of n copies of V). Indeed, any basis $v_1, \ldots, v_n$ of V gives rise to an isomorphism of representations $\mathrm{End}(V) \to nV$, given by $x \to (xv_1, \ldots, xv_n)$.

Remark 3.1.3. Note that by Schur's lemma, any semisimple finite dimensional representation V of A is canonically identified with $\bigoplus_X \mathrm{Hom}_A(X, V) \otimes X$, where X runs over all irreducible representations of A. Indeed, we have a natural map $f : \bigoplus_X \mathrm{Hom}(X, V) \otimes X \to V$, given by $g \otimes x \to g(x)$, $x \in X$, $g \in \mathrm{Hom}(X, V)$, and it is easy to verify that this map is an isomorphism. Indeed, if the result holds for representations V_i for $i \in I$, then it holds for their direct sum. Therefore one may assume that V is irreducible.